

HOW TO RUN — Easy Beginner Guide

This is the **plain-English, copy-paste guide** to run the Soccer Betting Toolkit on your own computer (Windows, Mac or Linux) and to connect the **live odds API**. No prior Python experience needed — just follow the steps in order.

 You only do **Part 1 (Setup)** once. After that you jump straight to **Part 3 (Run things)** whenever you want.

Part 0 — What you need first

1. **Python 3.9 or newer** installed.
 - Check by opening a terminal (see below) and typing: `python --version`
 - If you get an error, install it from <https://www.python.org/downloads/> (<https://www.python.org/downloads/>)
(on Windows, tick **“Add Python to PATH”** during install).
2. **The project folder** on your computer (this folder).

How to open a terminal:

- **Windows:** press `Win` , type **PowerShell**, hit Enter.
 - **Mac:** press `Cmd+Space` , type **Terminal**, hit Enter.
 - **Linux:** `Ctrl+Alt+T` .
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Part 1 — One-time setup

Copy these commands **one block at a time** into your terminal.

Step 1.1 — Go into the project folder

```
cd path/to/soccer-betting-models
```

Replace `path/to/` with the real location. Tip: type `cd` (with a space) then drag the folder into the terminal window and press Enter.

Step 1.2 — Create a private workspace (“virtual environment”)

This keeps the project’s libraries separate from the rest of your computer.

Windows (PowerShell):

```
python -m venv .venv
.venv\Scripts\Activate.ps1
```

Mac / Linux:

```
python3 -m venv .venv
source .venv/bin/activate
```

✓ You'll know it worked when you see `(.venv)` at the start of your terminal line.

Step 1.3 — Install the project and its libraries

```
pip install -r requirements.txt
pip install -e .
```

This downloads the math/ML libraries (numpy, pandas, scikit-learn, xgboost...).
It can take a few minutes the first time. ☕

Optional add-ons (only if you want them):

```
pip install -e ".[nn]"      # neural-network model (PyTorch)
pip install -e ".[dev]"    # to run the test suite
```

✓ **Setup done!**

Part 2 — Quick test (works offline, no API needed)

Make sure everything is wired up correctly by running the built-in demo on **fake (synthetic) data**:

```
python examples/quickstart.py
```

You should see it:

1. generate sample matches,
2. compare the models (Poisson, Dixon-Coles, etc.),
3. find "value bets",
4. run a backtest and print a summary.

If you see a **Backtest summary** table at the end, 🎉 it's working.

Part 3 — Running the real workflow

There are 4 simple commands. Run them in order. They all read their settings from `config/config.yaml` (you can edit that file to change leagues, seasons, etc.).

```
# 1) Download real historical results + odds (free, no key needed)
python scripts/collect_data.py
# ...or test offline with fake data:
python scripts/collect_data.py --synthetic

# 2) Train a model and see how good its forecasts are
python scripts/train_models.py --model dixon_coles
# other options: --model poisson | elo | ml

# 3) Backtest it (simulate betting on history with Kelly staking)
python scripts/run_backtest.py --model dixon_coles

# 4) List the value bets from a predictions file
python scripts/find_value_bets.py data/processed/predictions.csv --min-edge 0.03
```

The free historical data in step 1 comes from **football-data.co.uk** and needs **no API key**. You only need the API for upcoming/live matches — see Part 4.

Part 4 — Connecting the live Odds API

The free data above is **historical** (past matches). To get **upcoming matches and live odds**, the project uses **The Odds API**.

Step 4.1 — Get a free API key

1. Go to <https://the-odds-api.com/> (<https://the-odds-api.com/>)
2. Click **Get API Key** and sign up (the free tier gives 500 requests/month).
3. Copy your key — it looks like `a1b2c3d4e5f6...`

Step 4.2 — Give the key to the project

Never type your key inside the code. Instead, set it as an “environment variable” in your terminal:

Windows (PowerShell):

```
$env:ODDS_API_KEY = "paste_your_key_here"
```

Mac / Linux:

```
export ODDS_API_KEY="paste_your_key_here"
```

⚠ This lasts only for the current terminal window. To make it permanent, add that line to your shell profile (`~/.bashrc` , `~/.zshrc`) or use a `.env` file. The `.gitignore` already prevents keys from being uploaded to GitHub.

Step 4.3 — Turn the API on in the config

Open `config/config.yaml` and find the `odds_api` section. Change `enabled: false` to `enabled: true`:

```
data:
  odds_api:
    enabled: true          # ← was false
    base_url: https://api.the-odds-api.com/v4
    sport: soccer_epl      # league: soccer_epl, soccer_spain_la_liga, ...
    regions: eu            # eu | uk | us | au
    markets: h2h           # h2h = home/draw/away
```

Step 4.4 — Test the API connection in Python

Paste this into a file called `test_api.py` (or run `python` and paste line by line):

```
from soccer_betting.data.collectors import TheOddsAPICollector

# Reads your ODDS_API_KEY automatically from the environment.
collector = TheOddsAPICollector(sport="soccer_epl", regions="eu", markets="h2h")
odds = collector.fetch_odds()

print(f"Got {len(odds)} odds rows")
print(odds.head())
```

Run it:

```
python test_api.py
```

✓ If you see a table with `home_team`, `away_team`, `odds_home`, `odds_draw`, `odds_away`, your API is connected!

Common errors:

| Message | Fix |

|—|—|

| No Odds API key. Set ODDS_API_KEY... | You forgot Step 4.2 — set the key in the **same** terminal. |

| 401 Unauthorized | The key is wrong or has extra spaces/quotes. |

| 429 Too Many Requests | You hit the free monthly limit — wait or upgrade. |

| Empty table | No matches scheduled right now for that sport / regions. |

Part 5 — Use it from your own Python code

```
import pandas as pd
from soccer_betting.data.preprocess import make_synthetic_matches
from soccer_betting.models import DixonColesModel
from soccer_betting.evaluation.value import find_value_bets

# 1. Get some matches (here: synthetic; or load your collected CSV with pandas)
matches = make_synthetic_matches(n_teams=16, n_seasons=4)
train, test = matches.iloc[:700], matches.iloc[700:].reset_index(drop=True)

# 2. Train a model
model = DixonColesModel(xi=0.0018).fit(train)

# 3. Predict probabilities [home, draw, away]
preds = model.predict_frame(test).reset_index(drop=True)

# 4. Find value vs the bookmaker
frame = pd.concat(
    [test[["home_team", "away_team", "odds_home", "odds_draw", "odds_away"]], preds],
    axis=1,
)
print(find_value_bets(frame, min_edge=0.05, devig_method="shin").head())
```

Part 6 — Running the tests (optional, for peace of mind)

```
pip install -e ".[dev]"
pytest -q
```

All tests passing = the math (odds, Kelly, scoring, models) is behaving.

Quick troubleshooting

Problem	Likely fix
<code>python</code> not found	Try <code>python3</code> instead, or reinstall Python with “Add to PATH”.
<code>(.venv)</code> not showing	Re-run the activate command from Step 1.2.
<code>ModuleNotFoundError: soccer_betting</code>	Activate the venv, then re-run <code>pip install -e .</code>
<code>pip</code> is slow / fails	Check your internet; re-run the command (it resumes).
Permission error on Windows activate	Run PowerShell as Admin once: <code>Set-ExecutionPolicy RemoteSigned</code>

Cheat sheet (after setup)

```
# every new session:
cd path/to/soccer-betting-models
source .venv/bin/activate           # Windows: .venv\Scripts\Activate.ps1

# then do what you need:
python examples/quickstart.py           # offline demo
python scripts/collect_data.py          # get historical data
python scripts/train_models.py --model dixon_coles # train
python scripts/run_backtest.py --model dixon_coles # backtest

# for live odds, once per session:
export ODDS_API_KEY="your_key"          # Windows: $env:ODDS_API_KEY="your_key"
```

 **Reminder:** This toolkit is for research/education. Betting markets are efficient and edges are thin. Never stake money you can't afford to lose.

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